

Henry Chen

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<https://github.com/Chenry513>

TECHNICAL SKILLS

Languages: Python, C/C++ , SQL, Java
Machine Learning: TensorFlow, PyTorch, Keras, scikit-learn
Frameworks: Django, Flask, FastAPI, React, Node.js
Databases: PostgreSQL, SQLite
Tools: Git, GitHub Actions, Docker, VS Code

EDUCATION

Master of Science in Applied Computing British Columbia Institute of Technology — CGPA: 91%	Sept. 2025 – Dec. 2026 (est.) Burnaby, BC
Bachelor of Science in Computing Science Simon Fraser University	Sept. 2020 – Aug. 2025 Burnaby, BC

WORK EXPERIENCE

Mitacs Research Intern – Robotics & Computer Vision Seaspan Shipyards (BCIT Collaboration)	Jan. 2026 – Present Vancouver, BC
- Developing a Python-based data pipeline to collect and label welding image data for computer vision training. - Building and evaluating machine learning models to detect welding joints for robotic automation. - Collaborating with mechanical and robotics engineers to integrate vision models into real-world welding systems.	
Data Automation Undergraduate Research Assistant Simon Fraser University	May – Aug. 2025 Burnaby, BC
- Designed a Markdown-driven ingestion workflow for structured LLM data. - Implemented a Python-based linter using schema validation to enforce Django model constraints. - Integrated GitHub Actions CI to automatically validate and process submissions on each commit. - Prevented invalid entries from being deployed by enforcing schema compliance before publication.	
Backend Development Undergraduate Research Assistant Simon Fraser University	Aug. – Dec. 2024 Burnaby, BC
- Built and extended Django models and database schemas to store structured LLM data and user preferences. - Developed REST API endpoints using Django REST Framework to support data retrieval, user profiles, and authentication. - Customized the Django Admin interface with filters and visual indicators to streamline LLM data management. - Managed database migrations and validation logic to maintain data integrity across project updates.	

PROJECTS

Applied Data Science, BCIT 30-Day Hospital Readmission Prediction	Dec. 2025 github.com/Chenry513/COMP-9170-Final-Project
- Developed a patient-level 30-day hospital readmission risk model using 100K+ clinical encounters while preventing data leakage through GroupKFold cross-validation. - Built a reproducible ML pipeline with imbalance handling and leakage-free stacking for reliable model evaluation. - Evaluated performance using ROC-AUC, PR-AUC, and bootstrap confidence intervals to quantify uncertainty and prevent misleading performance claims in a healthcare setting.	
Affective Computing, SFU Emotion-Based Break Recommendation System	Apr. 2025 github.com/Chenry513/CMPT419_Final_project
- Built and evaluated two temporal deep learning pipelines (OpenFace+LSTM and CNN+LSTM) to detect frustration from gameplay video. - Modeled emotional dynamics over time rather than relying on frame-level classification.	

- Collected and manually annotated video clips (Anger vs Neutral), achieving 70% accuracy and $F1 = 0.70$.
- Deployed the trained model via a Flask web application enabling real-time video uploads and automated break recommendations.

Real-Time Web Application Personal Project
Rankr

July 2024
github.com/Chenry513/rankr

- Built a real-time polling system enabling groups to create, join, and rank shared options collaboratively.
- Implemented live updates using React, TypeScript, Socket.io, and Redis for low-latency synchronization.
- Designed backend logic to aggregate ranked votes and compute final group decisions.

Computer Vision, SFU
CIFAR-100 Image Classification

May 2023
github.com/Chenry513/PyTorch-Image-Classifier

- Designed and trained a convolutional neural network on the CIFAR-100 dataset (100 classes, 60K images), achieving competitive validation performance.
- Ranked 13th out of 120 students in a graduate-level computer vision Kaggle competition.
- Improved generalization using data augmentation and regularization to reduce overfitting.